

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.77567942687, median 1.25526036845, std: 1.84654076158
Reprojection error (cam1): mean 1.35297432055, median 0.795581790013, std: 1.69061985977
Gyroscope error (imu0): mean 4.27010274386, median 2.975359861, std: 4.2047905495
Accelerometer error (imu0): mean 7.04548817977, median 5.72617643882, std: 6.16689581844

Residuals

Reprojection error (cam0) [px]: mean 1.77567942687, median 1.25526036845, std: 1.84654076158
Reprojection error (cam1) [px]: mean 1.35297432055, median 0.795581790013, std: 1.69061985977
Gyroscope error (imu0) [rad/s]: mean 0.00187865221339, median 0.00130902386284, std: 0.00184991779976
Accelerometer error (imu0) [m/s^2]: mean 0.528609104625, median 0.429623742602, std: 0.462690049819

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.99822797 0.02215169 -0.05522874 -0.08402122]
 [0.05590497 0.03116766 -0.9979495 0.04979828]
 [-0.02038492 -0.99926867 -0.03235082 -0.20787115]
 [0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[-0.99822797 0.05590497 -0.02038492 -0.09089374]
 [0.02215169 0.03116766 -0.99926867 -0.20741001]
 [-0.05522874 -0.9979495 -0.03235082 0.03833098]
 [0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.011168208306

Transformation (cam1):

T_ci: (imu0 to cam1):
[[-0.99822046 0.02275596 -0.05511872 -0.1341399]
 [0.05580172 0.03055717 -0.99797416 0.04947435]
 [-0.02102559 -0.99927395 -0.03177261 -0.20925815]

```
[ 0.      0.      0.      1.    ]]
```

T_ic: (cam1 to imu0):

```
[[-0.99822046 0.05580172 -0.02102559 -0.14106172]  
 [ 0.02275596 0.03055717 -0.99927395 -0.20756553]  
 [-0.05511872 -0.99797416 -0.03177261 0.03533183]  
 [ 0.      0.      0.      1.    ]]
```

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.0143173002517

Baselines:

Baseline (cam0 to cam1):

```
[[ 0.99999981 -0.00009053 -0.00060754 -0.05024048]  
 [ 0.00009091 0.99999981 0.00061295 -0.00018887]  
 [ 0.00060748 -0.00061301 0.99999963 -0.00130551]  
 [ 0.      0.      0.      1.    ]]
```

baseline norm: 0.050257791789 [m]

Gravity vector in target coords: [m/s^2]
[0.97587801 -9.75568007 0.20686069]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [418.49092054825485, 419.01836890840974]

Principal point: [426.22309661597495, 240.62141826940754]

Distortion model: radtan

Distortion coefficients: [0.0018522553805433038, -0.004441289070573855, 0.002077622196671689, 0.0016072344697879144]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.088 [m]

Spacing 0.0264 [m]

cam1

Camera model: pinhole

Focal length: [418.1373181287808, 418.7047797794034]

Principal point: [426.89372915754194, 240.75318755797375]

Distortion model: radtan

Distortion coefficients: [0.0020101645129511725, -0.004634587177999576, 0.00212291561651336, 0.001783407038900539]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.088 [m]

Spacing 0.0264 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 484

Accelerometer:

Noise density: 0.00341036503941

Noise density (discrete): 0.0750280308669

Random walk: 0.00424360641439

Gyroscope:

Noise density: 1.99979456114e-05

Noise density (discrete): 0.000439954803451

Random walk: 9.23535168218e-08

T_i_b

[[1. 0. 0. 0.]

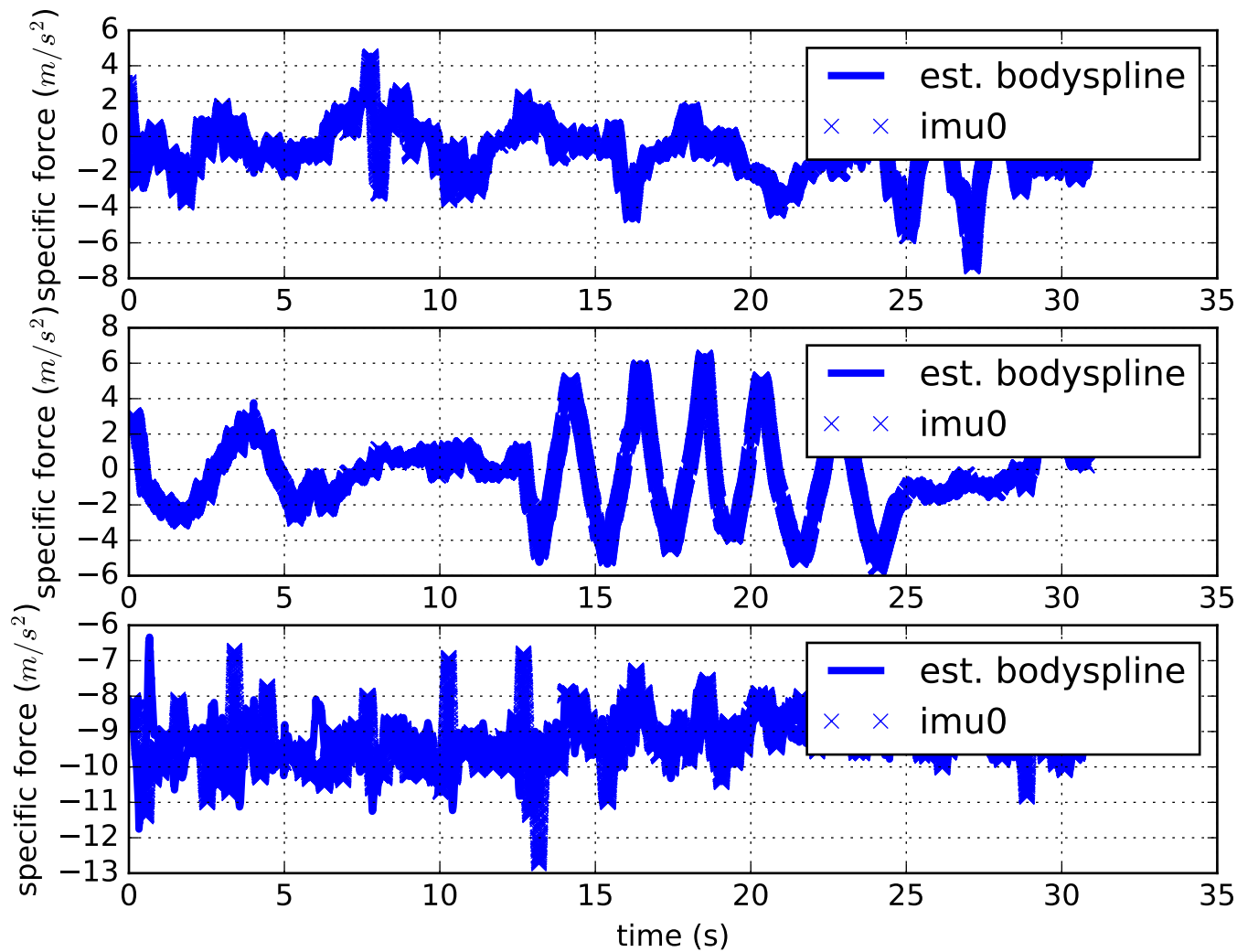
[0. 1. 0. 0.]

[0. 0. 1. 0.]

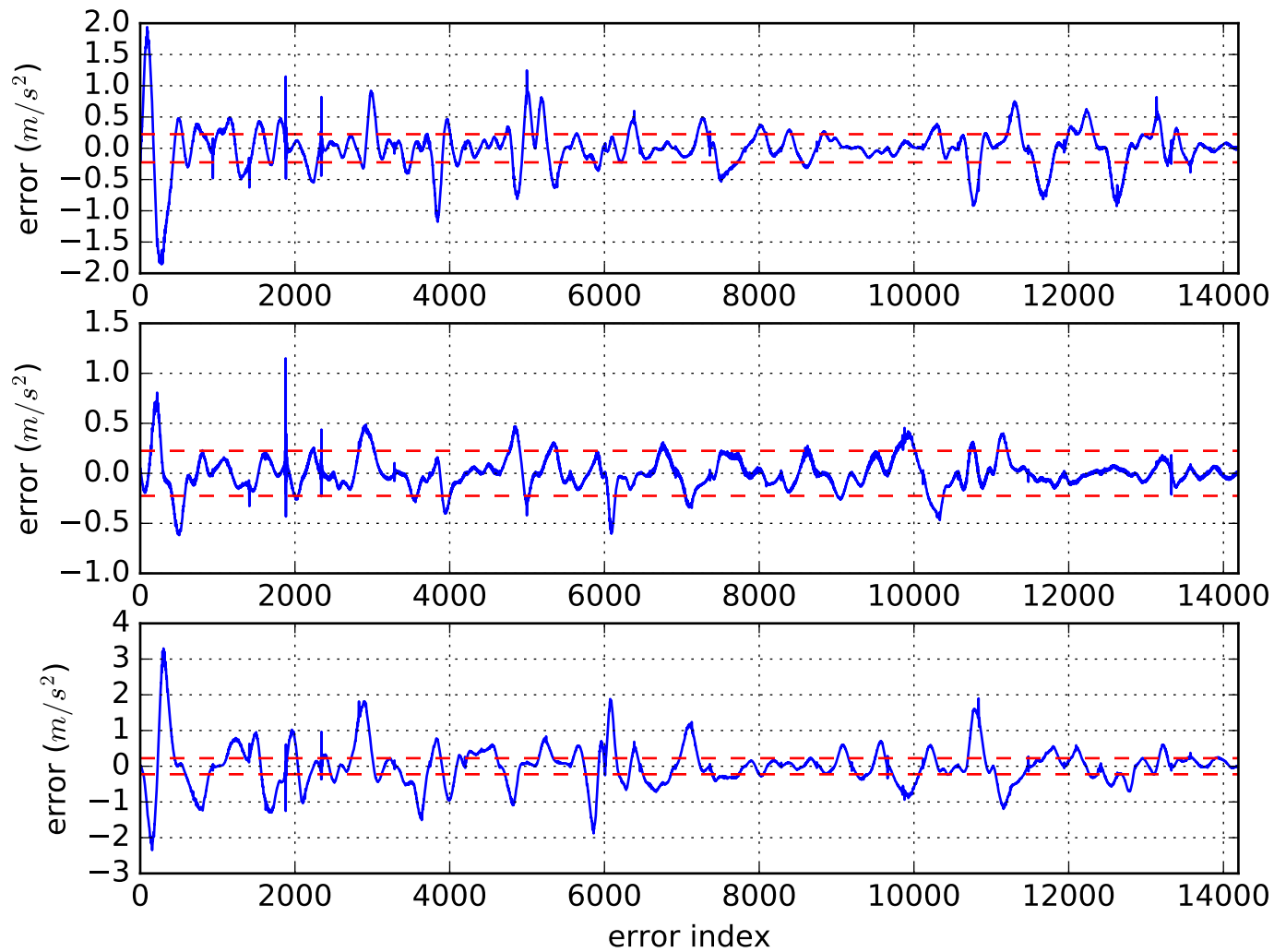
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

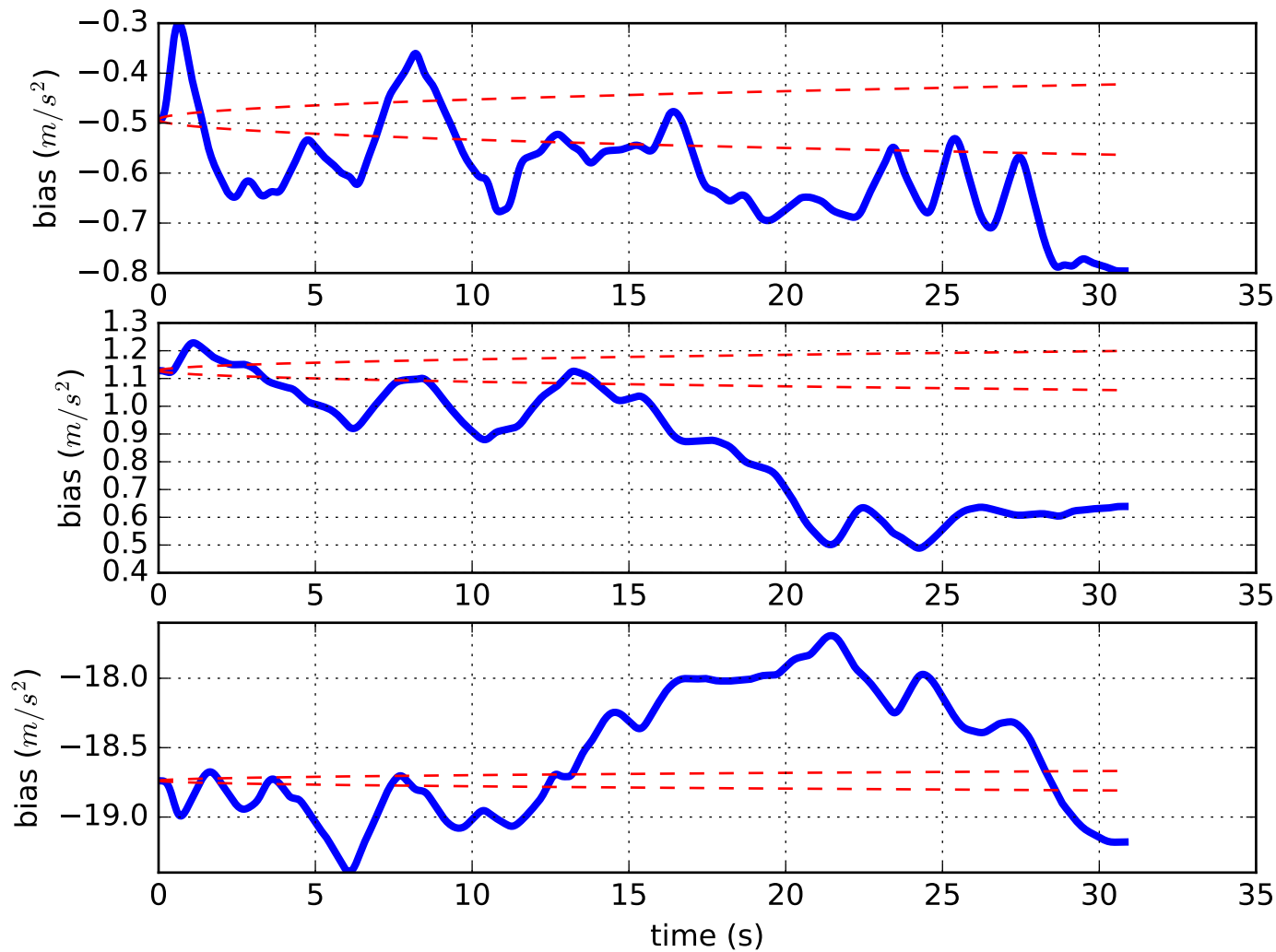
Comparison of predicted and measured specific force (imu0 frame)



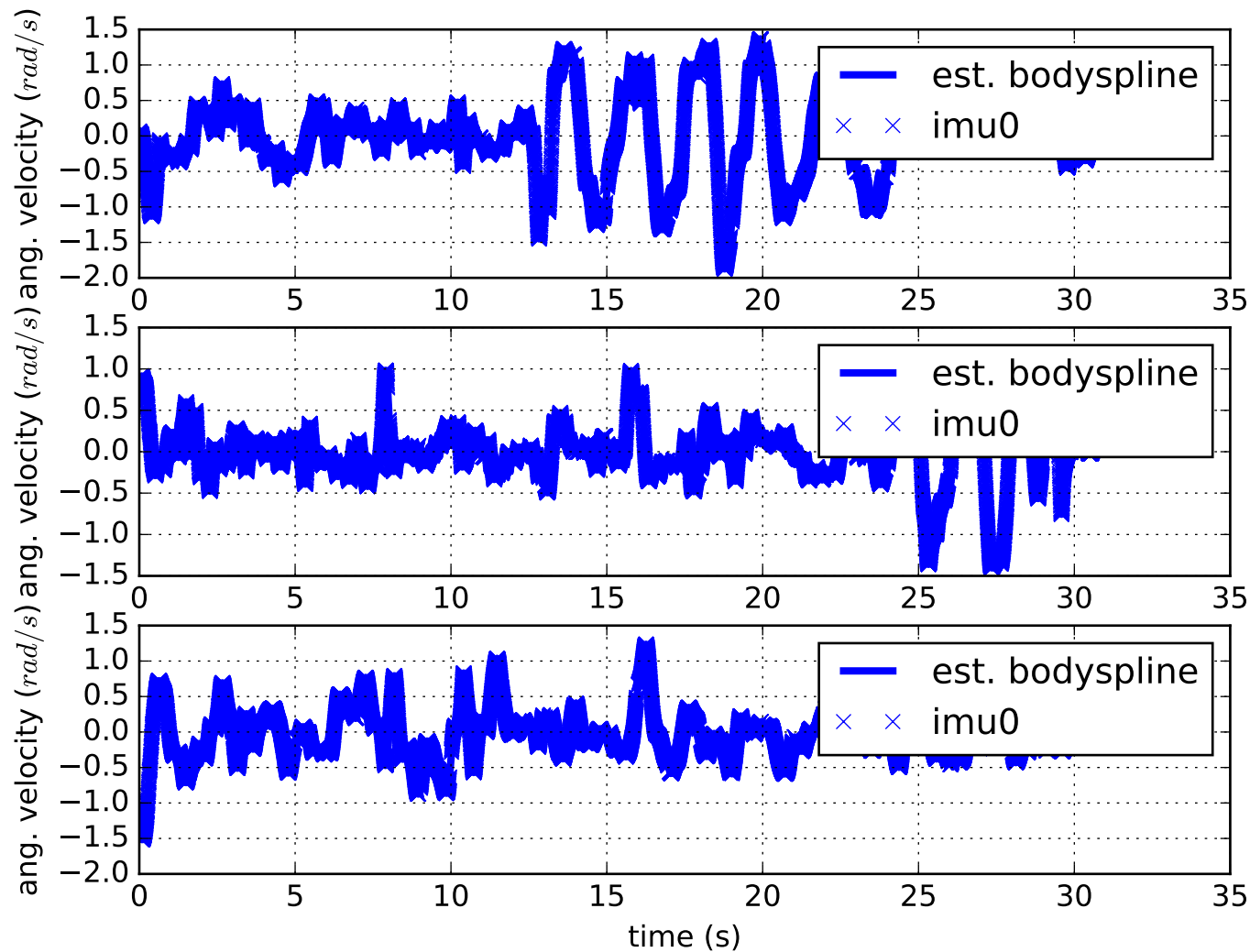
imu0: acceleration error



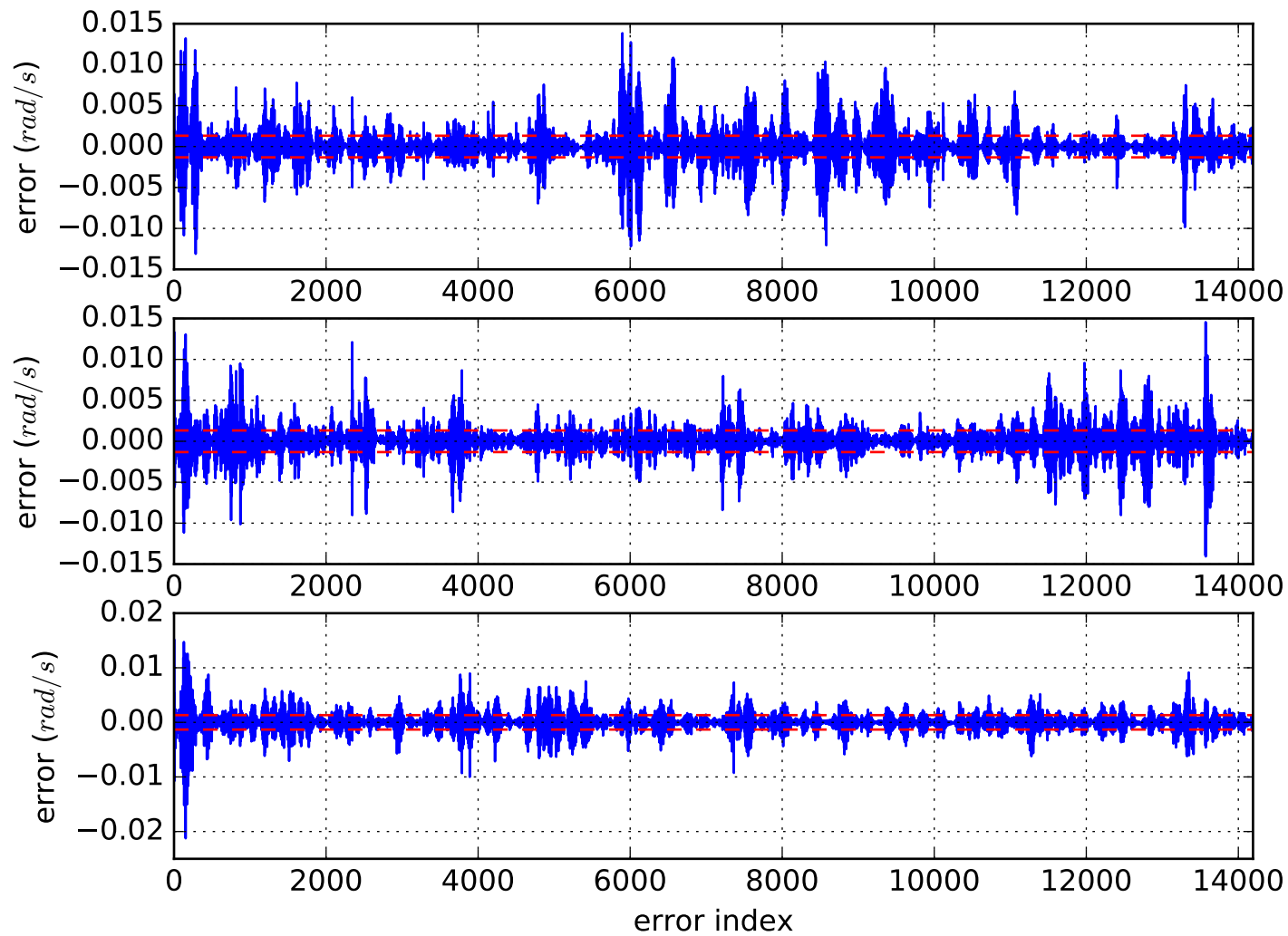
imu0: estimated accelerometer bias (imu frame)



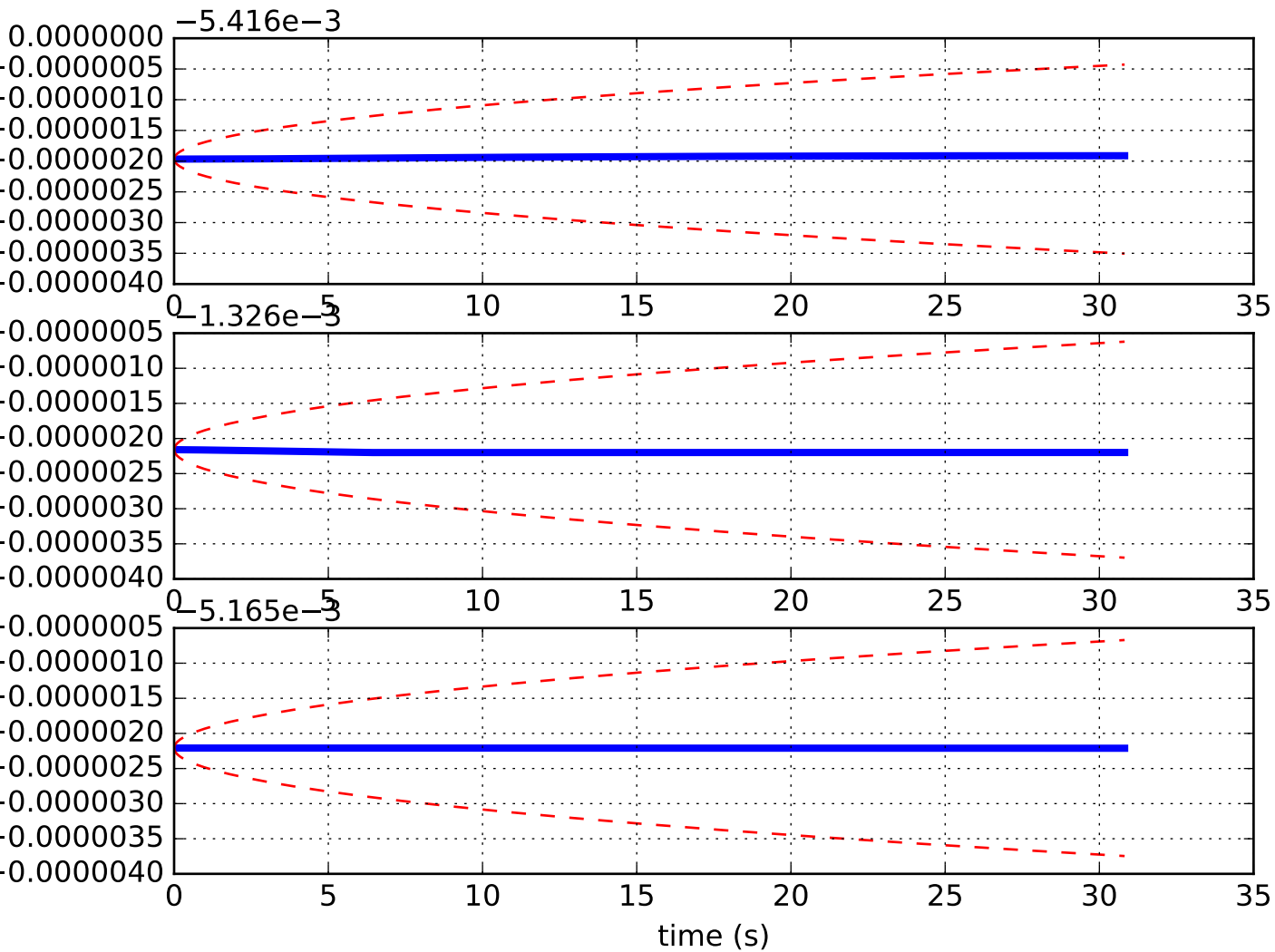
Comparison of predicted and measured angular velocities (body frame)



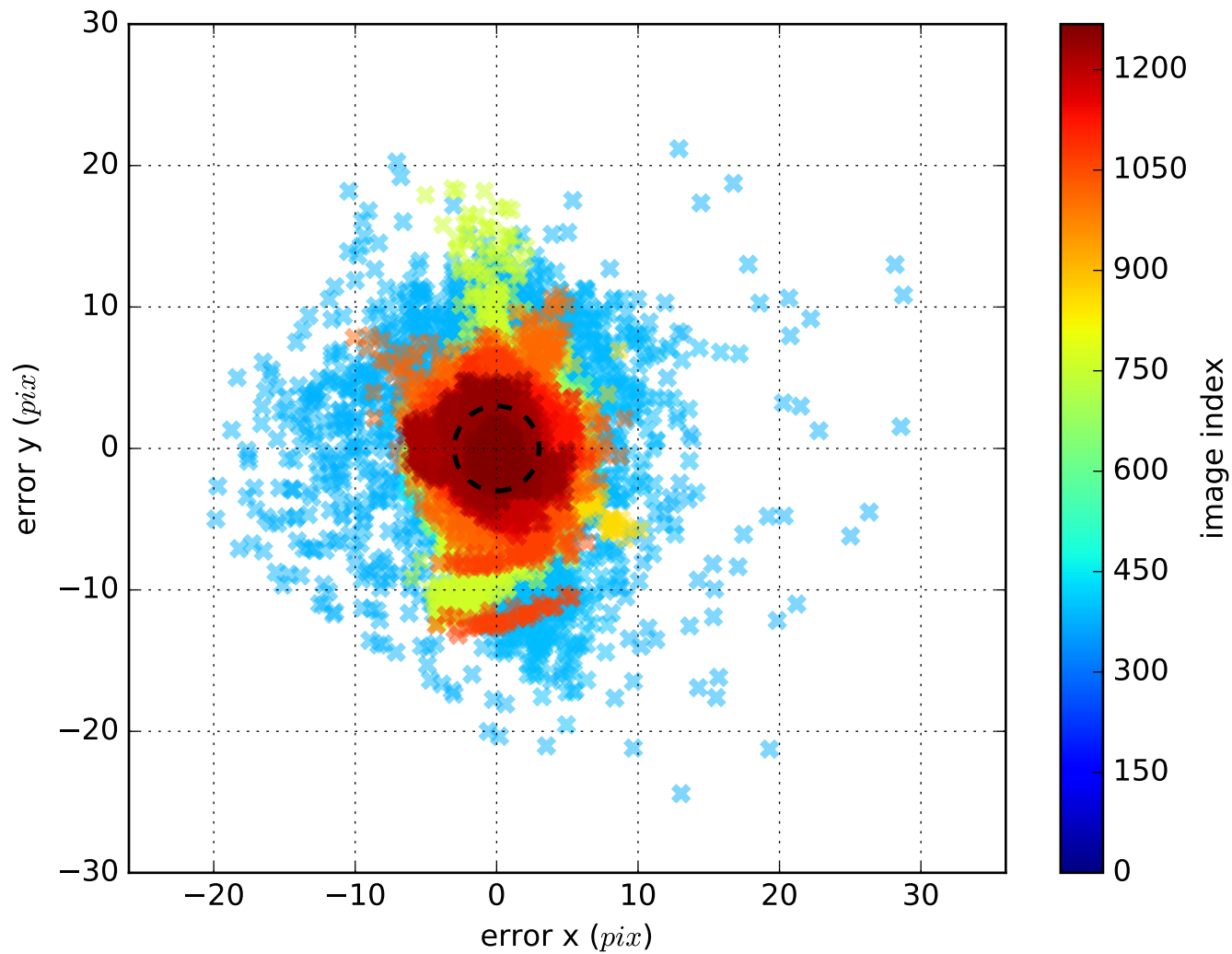
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

